OCR GCSE Computer Science
Paper 1: Computer Systems (Set A)

Tuesday 14 April 2026

Afternoon (Time: 1 hour 30 minutes)

Total Marks **/ 80**

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.

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1 (a) A computer system has a 2.5 GHz processor and 5 GB of RAM.

State the **purpose** of a CPU.

..... **(1 mark)**

(b) **State** what is meant by a **single core 2.5 GHz** processor.

.....

..... **(2 marks)**

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2 (a) **Identify** how many **unique values** can be represented by **4 bits**.

.....**(1 mark)**

(b) **Complete** the table by writing the answer to each statement

Statement	Answer
The smallest denary number that can be represented by a 4-bit binary number	
The largest denary number that can be represented by a 6-bit binary number	
The maximum number of different colours that can be represented with a colour depth of 7-bits	
The minimum number of bits needed to represent 150 different characters in a character set	

(4 marks)

(c) A musician uses a computer to make and record music.

Tick (✓) one box to identify the correct description of sound sampling.

A. The frequency of the wave is measured a set number of times each second.

B. The amplitude of the wave is measured at set intervals.

C. The digital sound wave is measured a set number of times each second.

D. The analogue sound wave's resolution is measured at set intervals.

(1 mark)

3 **Identify** and **describe** two **software-based security methods** that the company can use to protect their computer systems and data.

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.....

.....

..... **(6 marks)**

4 A technology company brings out new, updated devices twice a year.

Describe the environmental impacts of the company bringing out **new devices twice a year**.

.....

..... **(2 marks)**

5 (a) **Analyse** the potential consequences of a successful Distributed Denial of Service (**DDoS**) attack on a major e-commerce website and **explain** how this type of attack differs from a standard Denial of Service attack (**DoS**).

.....

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.....

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..... **(5 marks)**

(b) **State** the **definition** of the term '**social engineering**'.

.....

..... **(2 marks)**

(c) **Name two** other **threats** to the data in a computer system and **give a method** of **preventing** each.

.....

.....

.....

..... **(4 marks)**

(d) **Identify the threat** penetration testing can be used to help prevent.

..... **(1 mark)**

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6 (a) **Explain two disadvantages** to a law company of **storing their data** in the **Cloud**.

.....

.....

.....

..... **(4 marks)**

(b) The office introduces a WAP (**Wireless Access Point**) to allow network access to wireless devices.

The office manager has noticed that the **performance of the network has recently decreased**.

Describe how **introducing wireless** access could have **slowed down the network**.

.....

..... **(2 marks)**

(c) **Identify two** other factors that can **affect the performance** of a network.

.....

..... **(2 marks)**

(d) **Describe** the role of the switch in the star topology.

.....

.....

..... **(3 marks)**

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7 (a) Ali runs **defragmentation analysis** on his **magnetic hard disk**. Parts of the results are shown.

	File 1
	File 2
	File 3
	Free space

Explain how defragmentation will **change how the files and free space are arranged** on Ali's hard disk.

.....

.....

..... **(3 marks)**

(b) **After defragmentation**, Ali's computer **is able t**o **access files faster**.

Explain why Ali's computer can access the files faster after defragmentation.

.....

..... **(2 marks)**

(c) **Give three** additional **examples** of utility programs.

.....

.....

..... **(3 marks)**

8 (a) **State two functions** of the operating system.

.....

..... **(2 marks)**

(b) Ali's tablet computer has an operating system.

Complete the following description of the functions of an operating system **by selecting the appropriate missing words from those in the box**.

user, drivers, directories, hardware, interface, multitasking, output, peripherals, printers, processor, RAM, utility, ROM, running, passwords, faster, volatile, virtual.

The operating system provides a user ..... This displays the output to the user and allows the user to interact with the .....

The operating system controls the movement of data from secondary storage to .....and vice-versa. This is known as memory management.

The operating system can only perform one process at a time, but by managing the memory the computer can appear to be completing more than one process at a time. This is known as .....

An operating system allows device .....to be installed to allow an external piece of hardware to interact with the .....

The operating system provides security through user accounts and .....
It also creates and maintains a file system to organise files and .....

(8 marks)

9 (a) The smart television has **secondary storage**.

State, using **an example**, why the smart television needs **secondary storage**.

.....

..... **(2 marks)**

(b) Nina wants to transfer photos from a digital camera to an external **secondary storage** device.

Identify the **three** common **types of storage** Nina can choose from.

.....

.....

..... **(3 marks)**

(c) A musician is trying to decide if they should buy a magnetic secondary storage device or a solid state secondary storage device for their computer

Identify **one other** type of secondary storage.

..... **(1 mark)**

(d) Both computers have **RAM** and **ROM**.

The table has five statements describing RAM and/or ROM.

Tick (✓) one or more boxes in each row to identify if that statement describes RAM and/or ROM.

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	RAM	ROM
Stores data		
The memory is volatile		
Data will not be lost when the computer is turned off		
Data is read-only, cannot be changed		
Stores currently running data and instructions		

(5 marks)

(e) **Give one difference** between **RAM** and **flash memory**.

..... **(1 mark)**

(f) **State four characteristics** of **secondary storage devices** that Nina should consider when choosing a device.

.....

.....

.....

..... **(4 marks)**

10 (a) A library does not use encryption when data is transmitted through the network.

Give two reasons why the library **should use encryption**.

.....

..... **(2 marks)**

(b) The network has one **wireless access point** in the kitchen that transmits data on the **5GHz frequency**.

When the laptop is in the kitchen, it has better network performance.

Explain why the laptop's **network performance** is **lower in the bedroom.**

.....

..... **(2 marks)**

(c) **State two** ways Hope could **improve the wireless network performance** in the bedroom.

1	
2	

.....

..... **(2 marks)**